

WORK IN PROGRESS — draft syllabus subject to revision

COLLEGE OF ENGINEERING, UIC

CS 418, Introduction to Data Science, 3 Undergraduate Hours

I. Instructor & Course Details

Instructor Name

Email address:

Drop-In Office Hours (in-person or virtual):

Drop-In Hours location (*office number; link to Zoom, Google Hangouts, etc.*):

Co-Instructor/TA name

Email address:

Section designation(s):

Drop-In Office Hours (in-person or virtual):

Drop-In Hours location:

Blackboard Course Site LINK

Students are expected to log into the course site regularly to learn about any developments related to the course, upload assignments, and communicate with classmates. For all technical questions about Blackboard, email the Learning Technology Solutions team at LTS@uic.edu.

Course Modality and Schedule

II. Course Information

Catalog Course Description and Prerequisite/corequisite Statement

Growth Mindset:

Course materials and assignments can be complex and challenging, but they are crucial to your intellectual and personal growth and development. There are times you may need extra help.

Course Goals and Learning Outcomes

- Learning outcome 1
- Learning outcome 2
- Learning outcome 3

Course Topics

- Course introduction, statistics review, and the data science lifecycle
- Python foundations, dataframes, Polars, and obtaining data
- Data wrangling, filtering, and data formats
- Exploratory analysis and descriptive statistics
- Visualization basics, chart choice, visual encodings, visualization design, communication, and critique

- Hypothesis testing, estimation, sampling, and randomness
- Linear regression and classification
- Decision trees, support vector machines, and kernels
- Clustering, principal component analysis, and model evaluation
- Recommendation systems and A/B testing
- Network analysis, graphs, relationships, and network measures

III. Course Policies & Classroom Expectations

Grading Policy and Point Breakdown

Policy for Missed or Late Work

Attendance / Participation Policy

IV. Course Schedule

Weekly Schedule

Week	Class Day	Topic	Before Class	In Class
1	Monday	Course introduction; statistics review		
1	Wednesday	Data science lifecycle		
2	Monday	Python foundations; dataframes and Polars		
2	Wednesday	Obtaining data		
3	Monday	Wrangling data; filtering data		
3	Wednesday	Data formats		
4	Monday	Exploratory analysis		
4	Wednesday	Descriptive statistics		
5	Monday	Visualization		
5	Wednesday	Visualization		
6	Monday	Visualization		
6	Wednesday	Visualization		
7	Monday	Hypothesis testing		
7	Wednesday	Estimation and sampling; randomness		
8	Monday	Linear regression		
8	Wednesday	Intro to classification		
9	Monday	Decision trees		
9	Wednesday	Support vector machines; kernels		
10	Monday	Clustering; principal component analysis		
10	Wednesday	Model evaluation		
11	Monday	Recommendation systems		

Week	Class Day	Topic	Before Class	In Class
11	Wednesday	A/B testing		
12	Monday	Network analysis		
12	Wednesday	Graphs and relationships; network measures		
13	Monday	TBD		
13	Wednesday	TBD		
14	Monday	Presentations		
14	Wednesday	Presentations		
15	Monday	Presentations		
15	Wednesday	Presentations		

V. Accommodations

Disability Accommodation Procedures

Religious Accommodations

VI. Classroom Environment

Inclusive Community